BACE-1 Inhibitor Activity Prediction: Data Provenance, Integrity Audit, and Benchmark Tables

Yufan Tang · School of Pharmacy, Fudan University · 23307130372

NeurIPS-2026-format preview · every value computed from released result files

Table 1. BACE-1 dataset card. Source is the course-provided MoleculeNet split; the regression target is pIC_{50} (higher = stronger inhibition). RDKit canonicalization (once at ingest) and de-duplication leave the count unchanged.

Property	Value
Task	BACE-1 inhibition, regression (pIC_{50})
Source	BACE.split.csv (course-provided)
Molecules (total)	1,513
Train / Validation / Test	1,059 / 151 / 303 (predefined)
pIC_{50} mean ± std	6.52 ± 1.34
pIC_{50} range	[2.54, 10.52]
Canonicalization failures	0
Canonical duplicate SMILES	0
Undefined stereochemistry	1,078 / 1,513 (71%)
Protein structure (docking)	PDB 4D8C (public, RCSB; not in the CSV)

Table 2. Data-integrity audit of the predefined split. Top: train/test leakage diagnostics. Middle: a scaffold-disjoint re-split stress test (0% Murcko overlap) quantifies inflation from train/test similarity. Bottom: a label-permutation negative control; Pearson $R \approx 0$ confirms the signal is real, not memorized.

Diagnostic / test	Value	Note
Train/test leakage (predefined split)		
Canonical duplicate SMILES (train ∩ test)	0 (0.0%)	clean
Tautomer duplicates ($InChIKey_{14}$)	13 (4.3%)	disclosed
Murcko scaffold overlap (test in train)	193 (63.7%)	high
Nearest-train Tanimoto > 0.85	89 (29.4%)	high
Mean nearest-train Tanimoto	0.775	—
Scaffold re-split stress test (RF + Morgan, 3 seeds)		
Pearson R: predefined → scaffold	0.843 → 0.778	−0.065
MAE: predefined → scaffold	0.546 → 0.653	+0.107
Scaffold overlap: predefined → scaffold	64.0% → 0%	by design
Negative control (label permutation, 10 repeats)		
Permuted Pearson R	−0.06 ± 0.04	≈ 0, pass

Table 3. Five molecular representations on BACE pIC_{50} regression (predefined test split, mean ± std over 3 seeds). S1 = course strategy 1 (SMILES/fingerprint → ML); S2 = strategy 2 (graph → GNN); 3D = beyond syllabus. Best per column in **bold**. The graph network (Chemprop) wins on all four metrics.

Representation	MAE ↓	RMSE ↓	Pearson R ↑	Spearman ρ ↑
RF + Morgan [S1]	0.643 ±0.001	0.833 ±0.004	0.804 ±0.002	0.752 ±0.001
MolFormer-XL [S1]	0.657 ±0.076	0.859 ±0.095	0.798 ±0.045	0.746 ±0.046
ChemFM-3B [S1]	0.651 ±0.011	0.826 ±0.010	0.803 ±0.005	0.754 ±0.005
Chemprop (D-MPNN) [S2]	**0.604** ±0.017	**0.778** ±0.016	**0.835** ±0.007	**0.784** ±0.006

Table 4. Split-conformal prediction intervals on BACE (target coverage 0.90, mean ± std over 3 seeds; width in pIC_{50} units). The *tightest* interval (ChemFM) is not valid — it under-covers at 0.895 < 0.90 — whereas Chemprop gives the tightest *valid* interval (**bold**), ≈ ±1.33 pIC_{50}.

Representation	Coverage (target 0.90)	Width ↓	Status
RF + Morgan	0.934 ±0.003	2.873 ±0.018	valid
MolFormer-XL	0.924 ±0.009	2.934 ±0.310	valid
ChemFM-3B	0.895 ±0.007	2.633 ±0.050	under-covers (tightest)
Chemprop (D-MPNN)	0.925 ±0.011	**2.660** ±0.171	valid (tightest valid)

Table 5. Per-model training adequacy on BACE (3 seeds). Every model was trained to a validation plateau. RF has no epochs (budget = ensemble size). Chemprop uses the final epoch in production; a diagnostic re-run confirms epoch 50 sits on the plateau (final val loss within 0.005 of the best).

Representation	Selection rule	Budget	Selected epoch	Convergence (val)
RF + Morgan [S1]	ensemble size	500 trees	50 trees[†]	plateau by 50 trees
MolFormer-XL [S1]	best-on-val	30 ep	24.0 ±3.7	converged; high variance
ChemFM-3B [S1]	best-on-val	10 ep	7.7 ±1.2	clean plateau
Chemprop (D-MPNN) [S2]	final epoch	50 ep	50 (best 38)	final ≈ best (Δ 0.005)

[†] first tree count within 1% of the 500-tree validation RMSE.

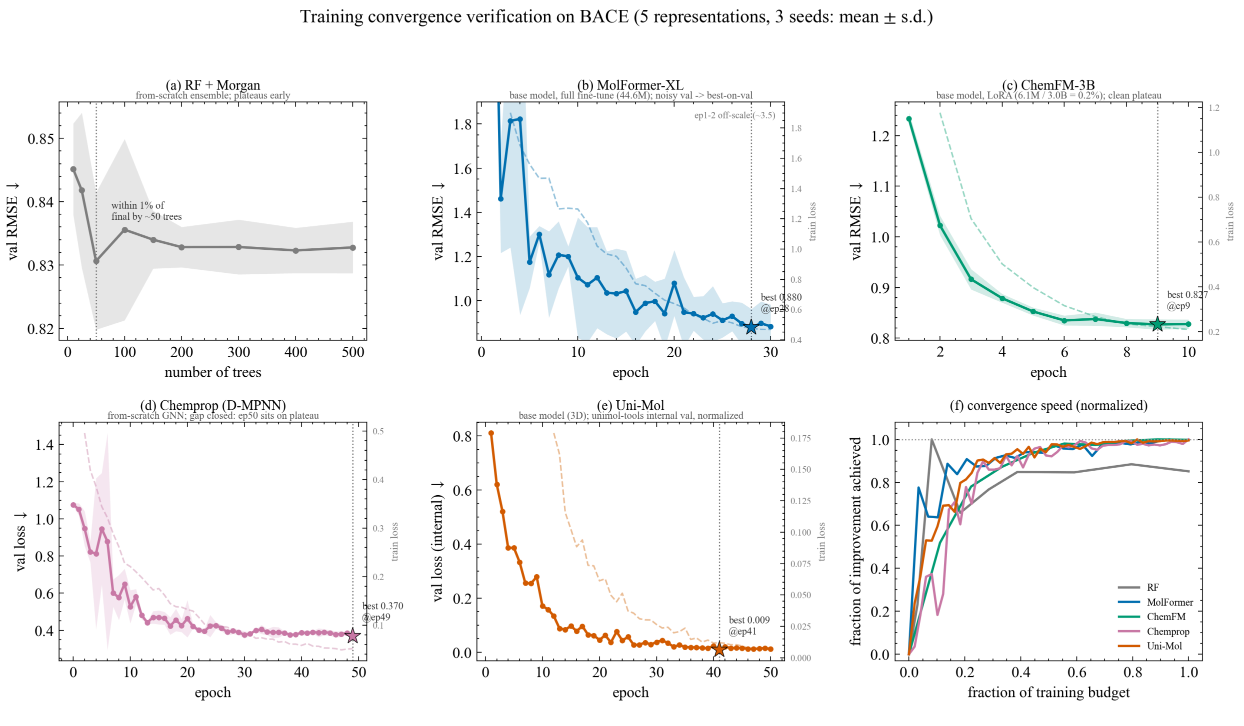

Figure 1. Training convergence verification on BACE (4 representations, 3 seeds; mean ± s.d.). Solid = validation trajectory with 3-seed band; faded = training loss (twin axis); ★ = selected epoch. (a) RF converges in tree count — flat past 50 trees; (b) MolFormer's wide band explains its largest seed variance; (c) ChemFM (LoRA, 0.2% of parameters) plateaus by epoch 8; (d) Chemprop's epoch-50 model is on the plateau (gap to best 0.005); (e) a normalized overlay compares convergence speed across all four.